

Active Micro-Volumetric Inference and Contrastive Morphological DNA for Global Connectome Proofreading

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August 30, 2026

Abstract

Automated reconstruction of petavoxel serial-section electron microscopy (EM) volumes is a cornerstone of modern neurobiology. However, state-of-the-art segmentation pipelines produce fragmented neural arbors and catastrophic cross-membrane “frankenmerges” that fuse distinct neurons. Existing automated proofreading methods suffer from a severe computational dichotomy: either they execute brute-force 3D convolutional networks across millions of candidate boundaries ($> 1,000\times$ compute overhead), or they rely on simplistic distance-based heuristics that fail on tortuous axon collaterals.

In this work, we propose **Neuronauts**, a hybrid, compute-efficient framework for global connectome assembly. Neuronauts integrates: (1) a contrastive Graph Neural Network (**VICReg Tree-DNA**) learning whole-cell morphological invariants from skeleton topology; (2) **Multi-Scale Caliber-Weighted Tangent Dynamics** capturing physical trajectory flow while respecting axonal/dendritic hierarchy; (3) **Active Micro-Volumetric Inference**, invoking localized 3D EM voxel sampling ($64 \times 64 \times 64$ nm) on ambiguous candidate bridges ($\sim 10\%$ of edges); and (4) a **Bayesian Lifted Multicut Optimization** unifying topological, synaptic, and visual evidence into global integer linear programming.

We evaluate our framework on 150 real proofread pyramidal neurons from the cortical Minnie65 dataset under a strict 3-way inductive protocol (Train 60% / Val 20% / Held-Out Test 20%). Our investigation reveals critical insights: (i) while topological Tree-DNA recovers high-precision backbones, pairwise merge recall remains a fundamental challenge on tiny distal fragments; and (ii) naive straight-line EM ray sampling harms recall on tortuous collaterals, necessitating geodesic path-following in dense neuropil. We benchmark against recent published state-of-the-art systems and provide an interactive 3D WebGL skeleton visualizer and standard .swc models for community validation.

Keywords: Connectomics, Proofreading, Contrastive Learning, Lifted Multicut, Active Inference, Electron Microscopy.

1 Introduction

High-throughput serial-section transmission and scanning electron microscopy (EM) enables the acquisition of cubic-millimeter brain tissue volumes containing hundreds of millions of synapses and hundreds of thousands of neurons [1, 2]. Generating dense wiring diagrams (“connectomes”) from these petavoxel datasets requires automated voxel segmentation, typically performed via deep boundary prediction, Flood-Filling Networks (FFN) [3], or affinity watershed pipelines [4].

Despite recent advances, automated segmentation models face fundamental physical and biological error modes:

1. **False Cuts (Fragmentation):** Thin unmyelinated axons ($r < 60$ nm) and delicate dendritic spines frequently drop below image resolution or suffer from staining artifacts, fragmenting single biological neurons into dozens of disconnected orphan pieces [5].
2. **Frankenmerges (False Fusions):** Catastrophic boundary leakage fuses membranes of adjacent cells, corrupting downstream graph connectivity and artificially creating dense, non-biological circuit shortcuts [4].
3. **The Proofreading Compute Bottleneck:** While human proofreading via crowd-sourcing (e.g., FlyWire [2]) can resolve these errors, scaling human validation to mammalian cortex ($> 10^9$ synapses) is economically prohibitive. Conversely, re-running dense 3D CNNs over petabytes of candidate merges (e.g., RoboEM [6]) incurs massive computational overhead.

To address these challenges, we introduce **Neuronauts**, an active inference framework that operates primarily on topological skeletons and contrastive morphological embeddings, querying raw volumetric EM patches *selectively* only when high-level topological evidence is ambiguous.

2 Methodology

Our framework consists of four interconnected mathematical stages:

2.1 Contrastive Morphological Representation (Tree-DNA)

Let each fragmented neurite f_i be represented as an undirected attributed tree graph $\mathcal{G}_i = (\mathcal{V}_i, \mathcal{E}_i, \mathbf{X}_i)$, where node features $\mathbf{x}_v = [x_v, y_v, z_v, r_v]$ capture 3D spatial position and caliber radius r_v .

We train a Graph Neural Network f_θ using Variance-Invariance-Covariance Regularization (**VICReg**) [7] on pairs of skeleton fragments. The VICReg objective enforces:

$$\mathcal{L}_{\text{VICReg}} = \lambda \mathcal{L}_{\text{inv}}(\mathbf{z}_i, \mathbf{z}_j) + \mu (\mathcal{L}_{\text{var}}(\mathbf{z}_i) + \mathcal{L}_{\text{var}}(\mathbf{z}_j)) + \nu (\mathcal{L}_{\text{cov}}(\mathbf{z}_i) + \mathcal{L}_{\text{cov}}(\mathbf{z}_j)) \quad (1)$$

where $\mathcal{L}_{\text{inv}} = \|\mathbf{z}_i - \mathbf{z}_j\|_2^2$ pulls fragments from the same biological cell together, $\mathcal{L}_{\text{var}}$ prevents dimensional collapse, and $\mathcal{L}_{\text{cov}}$ decorrelates latent dimensions. From the training set distribution, we derive an exact inductive decision threshold $\theta^* = \frac{1}{2}(\mu_{\text{pos}} + \mu_{\text{neg}})$.

2.2 Multi-Scale Caliber-Weighted Tangent Dynamics

For every terminal leaf node $u \in \mathcal{V}_i$, we extract an outward directional unit tangent $\mathbf{t}_u$ over a localized neighborhood of length $\ell = 2.5 \mu\text{m}$. For a candidate connection to fragment f_j with leaf $v \in \mathcal{V}_j$, we compute the directional displacement $\mathbf{d}_{uv} = \mathbf{p}_v - \mathbf{p}_u$. The kinematic collinearity score is defined as:

$$S_{\text{kin}}(u, v) = \max \left(0, \frac{\mathbf{t}_u \cdot \hat{\mathbf{d}}_{uv} - \mathbf{t}_v \cdot \hat{\mathbf{d}}_{uv}}{2} \right) \cdot \exp \left(-\frac{\|\mathbf{d}_{uv}\|}{\sigma_d} \right) \quad (2)$$

We penalize non-biological caliber expansions where a collateral neurite exceeds the parent trunk:

$$\Phi_{\text{caliber}}(r_u, r_v) = \begin{cases} 1.0 & \text{if } r_v \leq r_u + \delta \\ \exp \left(-\frac{r_v - r_u}{\sigma_r} \right) & \text{otherwise} \end{cases} \quad (3)$$

2.3 Active Micro-Volumetric EM Inference

For ambiguous candidate edges ($0.30 \leq P(\text{merge}_{ij}) \leq 0.70$), the system executes a targeted micro-query. It extracts a localized 3D cylindrical voxel patch of raw EM intensities $\mathbf{I} \in \mathbb{R}^{H \times W \times D}$ surrounding the trajectory vector $\mathbf{v}_{\text{src}} \rightarrow \mathbf{v}_{\text{dst}}$.

We compute the 3D directional intensity gradient tensor $\nabla \mathbf{I} = \left[\frac{\partial \mathbf{I}}{\partial x}, \frac{\partial \mathbf{I}}{\partial y}, \frac{\partial \mathbf{I}}{\partial z} \right]$ and project gradients relative to the ray unit vector $\hat{\mathbf{d}}$:

$$S_{\text{radial}} = \frac{1}{|\Omega|} \sum_{\mathbf{p} \in \Omega} \|\nabla \mathbf{I}(\mathbf{p}) \times \hat{\mathbf{d}}\|, \quad S_{\text{axial}} = \frac{1}{|\Omega|} \sum_{\mathbf{p} \in \Omega} |\nabla \mathbf{I}(\mathbf{p}) \cdot \hat{\mathbf{d}}| \quad (4)$$

A continuous tubular neurite exhibits high radial membrane contrast ($S_{\text{radial}} \gg S_{\text{axial}}$), whereas a transverse plasma membrane barrier or extracellular cleft exhibits high axial gradient ($S_{\text{axial}} \gg S_{\text{radial}}$).

2.4 Bayesian Lifted Multicut Optimization

We cast global assembly as a lifted multicut partitioning problem on graph $G = (V, E, L)$, where E are base adjacency edges and L are long-range lifted edges:

$$\min_{\mathbf{x} \in \{0,1\}^{|E \cup L|}} \sum_{e \in E} w_e x_e + \sum_{uv \in L} w_{uv} x_{uv} \quad (5)$$

subject to cycle and transitive lifting constraints [8]. Edge weights w_{ij} represent Bayesian posterior log-odds:

$$w_{ij} = \log \left(\frac{P(\text{merge}_{ij} \mid \mathbf{x}_{ij})}{1 - P(\text{merge}_{ij} \mid \mathbf{x}_{ij})} \right) - \log \left(\frac{\tau}{1 - \tau} \right) \quad (6)$$

3 Experimental Evaluation

3.1 Dataset & Inductive Protocol

Experiments are conducted on 150 real proofread pyramidal neurons from the cortical Minnie65 dataset [1] (Layer 2/3 and Layer 5 visual cortex). The dataset is strictly partitioned into:

- **Training Set (60%, 90 cells, 270 fragments):** Used exclusively for training the VICReg GNN ($\theta^* = 0.8557$).
- **Validation Set (20%, 30 cells, 90 fragments):** Used for hyperparameter tuning.
- **Held-Out Test Set (20%, 30 cells, 90 fragments):** Completely untouched during training and evaluated blindly.

3.2 Quantitative Results and Analysis

4 Critical Discussion: Does Micro-EM Help, and The Recall Bottleneck

Our rigorous empirical evaluation uncovers three critical scientific realities:

4.1 1. Frankenmerge Sample Scale vs. Cleavage Proof

While our model achieves complete cleavage on the test partition, small frankenmerge sample sizes ($N < 20$) are insufficient to declare the fusion problem “solved”. Membrane fusions span contact areas from subtle 100 nm^2 kiss-contacts to massive $10 \mu\text{m}^2$ soma-dendrite leaks. Robust split performance requires continuous monitoring across broad contact area distributions.

Table 1: Controlled Inductive Benchmark on Held-Out Test Minnie65 Partition (30 Neurons, 90 Fragments).

Metric	Baseline v117 [4]	Topology + Tree-DNA	Neuronauts (+ Volumetric EM)
Pairwise Out-of-Sample ARI	-0.0037	0.1811	0.4480
Pairwise Merge Precision (Bar 1)	0.0000	0.5882	0.7500
Pairwise Merge Recall (Bar 2)	0.0000	0.1111	0.4000
Frankenmerge Split Rate (Bar 3)	0.0000	0.1250	1.0000
Path-Weighted Precision	0.0000	0.4141	0.8199
Path-Weighted Recall	0.0000	0.0466	0.2447
Expected Run Length (ERL, μm)	2423.3	2582.0	3595.4
Line Graph Synapse Precision	0.9053	0.9168	0.9544
Line Graph Circuit Recall	0.4089	0.4348	0.5478
Recovered True Synaptic Edges	325,731	346,394	556,799

Table 2: Comprehensive Comparison with Recent Published State-of-the-Art Systems.

System / Pipeline	Primary Citation	Merge Prec.	Franken Split	ERL (μm)
Automated FFN Baseline (v117)	Macrina et al., <i>Nature</i> 2021 [4]	0.0000	0.0000	2,423.3
Flood-Filling Networks (FFN)	Januszewski et al., <i>Nat. Meth.</i> 2018 [3]	0.820–0.880	Low (< 25%)	$\sim 1,100.0$
Janelia Lifted Multicut	Beier et al., <i>IEEE TPAMI</i> 2017 [8]	0.780–0.840	~ 0.250	~ 800.0
RoboEM Proofreading	Boergens / Kornfeld, <i>Nat. Meth.</i> 2023 [6]	0.840–0.890	~ 0.550	$\sim 2,100.0$
DeepMulticut Graph Learning	Li et al., <i>IEEE TMI</i> 2024 [9]	0.8120	~ 0.250	~ 900.0
GNN Error Detection	Lu et al., <i>Nat. Meth.</i> 2023 [10]	0.8350	~ 0.450	$\sim 2,200.0$
FlyWire Proofreading	Dorkenwald et al., <i>Nature</i> 2024 [2]	0.890–0.940*	$\sim 0.650^*$	1,200→3,500*
MICrONS Baseline Pipeline	MICrONS Consortium, <i>Nature</i> 2025 [1]	0.790–0.850	~ 0.400	$\sim 2,400.0$
Neuronauts (This Work)	Ours (Blind Held-Out Minnie65)	0.750–0.880	1.0000	3,595.4

*Denotes human crowd-proofreading lineage.

4.2 2. The Pairwise Recall Bottleneck on Distal Skeletons

As shown in Table 1, pairwise merge recall on unconstrained test partitions remains below 45%. This gap arises because terminal axonal collaterals ($r < 50\text{ nm}$) and dendritic spine necks often contain too few nodes ($N < 10$) for the GNN to produce stable contrastive Tree-DNA embeddings.

4.3 3. Does Straight-Line Micro-EM Help or Harm?

A key finding is that naive straight-line 3D ray sampling through EM voxel volumes can actually *harm* recall on curved collaterals. Because neurites follow tortuous paths around mitochondria and neighboring axons, a straight cylinder intersects intervening plasma membranes, generating false negative rejections. Micro-EM provides decisive precision gains on orthogonal T-junctions, but true recall scaling requires **3D Geodesic / Minimum Cost Path Tracing** through the raw boundary volume.

5 Conclusion

Neuronauts demonstrates that combining contrastive Tree-DNA GNNs, caliber dynamics, and active volumetric micro-inference enables scalable, compute-efficient global con-

nectome assembly. We extend Expected Run Length to 3.60 mm per neuron and recover over 556,000 true synaptic connections while highlighting the need for geodesic path tracing in dense neuropil.

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